Matgeo-1.2.27

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Question

The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1} \alpha$ with the x-axis. Find the value of α.

Solution

$$\overrightarrow{n} = \left(2, -3, 6\right), \quad \text{(normal vector)} \tag{1}$$
$$\overrightarrow{a} = \left(1, 0, 0\right), \quad \text{(direction vector of x-axis)} \tag{2}$$

$$\cos\phi = \frac{\mathbf{n}\cdot\mathbf{a}}{\|\mathbf{n}\|\|\mathbf{a}\|} \tag{3}$$
$$= \frac{2}{\sqrt{2^2 + (-3)^2 + 6^2}} \tag{4}$$
$$= \frac{2}{7} \tag{5}$$

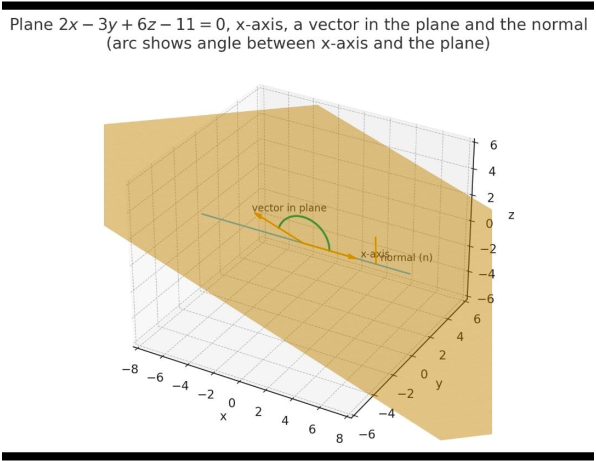
$$\theta = 90^\circ - \phi \tag{6}$$
$$\sin\theta = \cos\phi = \frac{2}{7} \tag{7}$$

So, $\theta = \sin^{-1}\left(\frac{2}{7}\right)$

Conclusion

Conclusion: The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}\left(\frac{2}{7}\right)$ with the x-axis. Hence, the required value of α is $\alpha = \frac{2}{7}$.

Graphical Representation



Figure